

Week 3 Assignment Solutions

Question 1: BFS Node Inspection Order Answer: S,B,E,J,O,A,F

Question 2: Node with largest heuristic value (Manhattan distance to G) Answer: B,16 (or O,16)

Question 3: Best First Search Inspection Order Answer: S,J,I,E,F,K,M

Question 4: Hill Climbing node sequence Answer: S,J,I

Question 5: Hill Climbing node sequence (Extended) Answer: S,J,I,J,I,J,I

Question 6: Local Optimum Node Answer: I

Question 7: Modified Question Path Answer: S,B,A,C,D,G

Question 8: Number of Hamiltonian Cycles Answer: 22

Question 9: Nearest Neighbour Heuristic Answer: A. Nearest Neighbour algorithm fails to find a tour

Question 10: Greedy Heuristic Answer: A. Greedy algorithm fails to find a tour

Question 11: SAT Heuristic Values for S, D, E Answer: 2,3,4

Question 12: Beam Search (w=2) Level 2 Nodes Answer: 00101,10001

Question 13: Beam Search (w=2) Level 3 Node Heuristic Values Answer: 5,5

Question 14: Exact Variable Assignment for SAT formula Answer: 00110

Question 15: Select the correct statements Answer: A, D

Question 16: Complete algorithms for finite state/solution space Answer: A, B, C

Question 17: Working with 2-City-Exchange operator Answer: C, D

Question 18: Iterated Hill Climbing algorithm Answer: B, E

Question 19: Why Hill Climbing may run into a local optimum Answer: C

Question 20: Variable Neighbourhood Descent Answer: A, C